

Outlet Intelligence

Unmasking latent demand for 20,000 Sri Lankan beverage outlets

OUTLETS PREDICTED

20,000

HOLD-OUT SPEARMAN ρ

0.793

LKR ALLOCATED

4,999,913

COOLER TOP-100 NET

LKR 31.65 M

The hidden half of demand

$$V = \min(D, C)$$

observed = min (latent demand , systemic constraint)

WHAT WE OBSERVE

Historical V — every distributor invoice. Whatever the outlet actually bought. Constrained, censored, biased downward whenever credit, cooler, or supply got in the way.

WHAT WE NEED

Latent D — what a healthy version of the outlet would buy in January 2026 if credit, cooler and supply constraints were relaxed. That is the number that should drive trade-spend and cooler-deployment decisions.

Five textbook methods converge on one answer

Peer Q90	Log-linear	Tobit MLE	Stochastic Frontier	Phase-3 formula
ANCHOR	CROSS-CHECK	CROSS-CHECK	CROSS-CHECK	CROSS-CHECK
Cohort-conditional 90th percentile	OLS on unconstrained subset	Tobin 1958 censored likelihood	Aigner-Lovell-Schmidt half-normal	Interpretable business benchmark
weight 0.40	weight 0.18	weight 0.18	weight 0.16	weight 0.08

Pairwise Spearman ρ — all five rank the outlets the same way

peer vs log-linear 0.86 · peer vs Tobit 0.84 · peer vs SFA 0.85 · peer vs Phase-3 0.84 · log-linear vs Tobit 0.92

No supervised ML. Training on V would reproduce V 's censoring. Five independent textbook methods converge on the same ranking.

43,000 POIs — distance-decay, not flat counts

9 SEMANTIC CATEGORIES

Footfall

σ 150m

School

σ 400m

Tourist

σ 1km

Health

σ 400m

Worship

σ 300m

Food service

σ 150m

Leisure & rec

σ 500m

Population

σ 600m

Competitor

gravity

MARKET STRUCTURE (NON-LINEAR)

HHI · WESTERN PROVINCE (fragmented)

1,101

HHI · NORTH-WESTERN (concentrated)

4,319

TYPE-WEIGHTED COMPETITION PRESSURE

Grocery × Grocery = 1.0

A bus stop 20 m away is not the same as one 400 m away. Gaussian decay for demand drivers, gravity decay for competitors.

How well does it actually generalise?

HOLD-OUT SPEARMAN ρ (Jan 2025 predicted from 2023-24 only)

0.793

over 10,949 evaluable outlets

SUPPORTING EVIDENCE

Pairwise method agreement

all $\rho \geq 0.83$

5-method ensemble triggered

XAI surrogate fidelity

ρ 0.97 · R^2 0.998

SHAP attributions are faithful

Bit-exact reproducibility

verified

Fresh-clone produces identical CSV

Sanity assertions on output

all passed

Floors/ceilings respected

Hold-out validation pretends Jan 2025 doesn't exist. The rank correlation against actuals is point seven nine.

Where should Western Province spend the budget?

CONCAVE WATER-FILLING LP

maximise $\sum \kappa_i v(x_i)$ subject to $\sum x_i \leq 5,000,000 \text{ LKR}$ · $0 \leq x_i \leq 5,000 \text{ LKR per outlet}$

$$\kappa_i = \text{uplift} \times \text{cooler_priority} \times (1 + 0.4 \text{ spatial}) / (1 + 0.3 \text{ competition})$$

DISCOUNT

LKR 1.74 M

share of LKR 5 M · 34.9 %

MERCHANDISING

LKR 2.09 M

share of LKR 5 M · 41.8 %

PROMOTIONAL

LKR 1.17 M

share of LKR 5 M · 23.3 %

LKR 4,999,912.87 allocated across 8,416 funded outlets (172 zero-uplift outlets correctly excluded).

The single strongest recommendation — hard money

NET 24-MONTH VALUE

LKR 31.65 M

from a LKR 5 M cooler programme · median payback 3.33 months

Metric	Value
Capex required (100 × LKR 50,000)	LKR 5,000,000
24-month margin uplift (predicted)	LKR 36,650,325
Net 24-month value	LKR 31,650,325
Median payback period	3.33 months
Outlets currently without a cooler (eligible)	~6,500

Live demo · 5 minutes · <http://localhost:3000>

1

DASHBOARD

Six KPI tiles
Province breakdown
Quick links

2

OUTLETS

20,000 outlets
Six filters
20 per page

3

DRILL-DOWN

Predicted potential
SHAP drivers
What-if deltas
Recommended actions

4

AI EXPLANATION

Gemini 2.5 Flash
Grounded in payload
Plain-English narrative

INSIGHTS WORKSPACE — TABBED VIEWS

Budget · Cooler ROI · Dormancy risk · Distributor scorecard · Territories (Leaflet map) · Forensics findings

From this repo to the field in 90 days

MONTH 1

PILOT

- Deploy 30 of the Top-100 coolers across two distributors
- Track weekly uplift vs predicted
- Train sales reps on the Outlet Intelligence app
- Deploy silver-clean DQ checks into the live SFA/ERP pipeline

MONTH 2

CALIBRATE

- Fit β (litres-per-LKR elasticity) from pilot data
- Operationalise weekly dormancy alerts
- First distributor scorecard review
- Audit forensic findings in production

MONTH 3

SCALE

- Deploy remaining 70 coolers in the Top-100
- A/B test discount vs merchandising on 200 outlets/channel
- Re-baseline framework with 2026 actuals
- First quarterly trade-spend review driven by this system

What this delivers

Capital efficiency

LKR 31.65 M net 24-month value from a LKR 5 M cooler programme

Sales coverage

Top-200 at-risk outlets flagged before they lapse · AUC 0.879

Channel discipline

First per-outlet split of trade spend (34.9 / 41.8 / 23.3)

Operational visibility

10-distributor scorecard · two-standard-deviation best-to-worst gap

Decision integrity

Every recommendation traceable to SHAP driver + counterfactual scenario

The dashboard is live. The Top-100 list is ready. The first cooler can go out next week.